Lab 2

4-bit Binary Adder and Subtractor

1. Introduction

In this lab you will design a circuit capable of adding and subtracting two 4-bit binary numbers. The circuit will have an overflow indicator for both signed and unsigned numbers addition. You will use the Quartus Prime Schematic to implement it. You will upload your design into the FPGA on the DE1-SoC Board and verify the design using Switches and LEDs on the board. You are **not** allowed to use any of the Quartus software library modules unless for the modules that are explicitly allowed in the lab instructions.

1.1 Prelab Assignment

What is the difference between overflow in adding two signed binary numbers vs adding two unsigned binary numbers? Give numerical examples to support your answer.

Have the following files from the previous lab assignment ready to be included in the design of this lab:

- **half_adder.bsf**
- **half_adder.bdf**
- **full_adder.bsf**
- **full_adder.bdf**

Have the requirements of the prelab assignment ready to be checked at the beginning of the lab session.

Unsigned Overflow: Occurs when the result exceeds the range [0, 15] for 4-bit numbers

Example: 15 + 1 = 16, but 16 cannot be represented in 4 bits

Signed Overflow (Carry into the MSB not equal to Carry out): Occurs when the result exceeds the range [-8, 7] for 4-bit 2's complement

Example: 7 + 1 = 8, but 8 cannot be represented in 4-bit 2's complement
Example: (-8) + (-1) = -9, but -9 cannot be represented in 4-bit 2's complement

2. The 4-Bit Adder Circuit

The required 4-bit adder is a circuit capable of adding two 4-bit binary integers that are either signed or unsigned numbers. The circuit will have 9 inputs, and 5 outputs as follows:

- 4 inputs for each of the two 4-bit numbers (**A[3]..A[0]** and **B[3]..B[0]**) using 8 switches.
- One input, using switch number 9, to indicate the sign of the input numbers (0 for unsinged integers and 1 for signed integers).
- 4 outputs for **Sum (S[3]..S[0])** using 4 LEDs
- One output for the **Overflow** using the last LED (LED number 9).

Carry out the following tasks:

a) Create a new folder for a new project named **four_bit_adder** and copy the **half_adder.bsf**, **half_adder.bdf**, **full_adder.bsf** and **full_adder.bdf** files from your previous projects folders to this new project folder.

b) Build a 4-bit adder schematic with **four** of the **full adders**.
Note: you can set an input to be always zero or always one by connecting it to **GND** or **VCC** respectively (search for them in the *Symbol Tool*).

c) Add necessary logic to your four_bit_adder to detect **Overflow**. Your overflow circuit must act based on the sign of the input numbers, signed or unsigned, given that the singed numbers are represented in 2's complement. You will need to prepare a truth table for this logic and minimize it using a Karnaugh map.

d) Assign the first 4 switches to inputs (**A[3]..A[0]**) and assign the following 4 switches to inputs (**B[3]..B[0]**). The last switch on the board, **SW[9]**, must be used as the signed/unsigned selector.

e) Assign the 4-bit **Sum** output to the first four LEDs and the overflow bit to the last LED (LED number 9, **LEDR[9]**).

f) Validate your design on the DE1-SoC board with different inputs. You must try some inputs that generate overflow.

Assignment 1
Design and implement the 4-bit Adder in Quartus schematic. Include the following in your report:

- Screenshot of your design.
- The truth table used to come up with the logic needed to detect the overflow output.
- Pictures of the board showing at least 8 test inputs that must include the following cases:
  - Two unsinged numbers with and without overflow.
  - All combinations of signed numbers (two positive, two negative, one positive/one negative) with and without overflow.

Verify the outputs of your circuit (by converting all input/output numbers to decimal) and explain the results.

Note: to avoid a large report file, make sure to reduce the resolution of the pictures by using a picture management tool that allow saving pictures for print purposes.

3. The 4-Bit Adder/Subtractor Circuit

Here you need to modify the previous 4-bit adder circuit to also handle the subtract operation. For this circuit assume both 4-bit input numbers are always **signed integers** (no need to handle unsigned numbers). Therefore, you do not need the input to indicate the sign of the numbers. Instead, that input will be used to indicate the needed operation (Add or Subtract). We will also refer to the output as **Rslt**, instead of **Sum.**

Let's start creating our new **four_bit_add_sub** project as a copy of the previous **four_bit_adder** project using the following steps:

- Open the **four_bit_adder** project in Quartus then select **Project → Copy Project**.
- Specify the destination new directory and new project name as **four_bit_add_sub**.
- Make sure that "**Open new project**" option is checked.
- Click **Yes** to create the new project directory.
- You will notice that the top level **bdf** file remains as **four_bit_adder.bdf**. To change that to the new **four_bit_add_sub.bdf** do the following:
  - Make sure that **four_bit_adder.bdf** is open in Quartus, then go **File → Save As →** save it in the new project directory as **four_bit_add_sub.bdf**
  - Go to **Project → Add Current File to Project**
  - Go to **Project → Set as Top-Level Entity**
  - Go to **Project → Add/Remove Files in Project →** select **four_bit_adder.bdf →** click **Remove → Ok**
- To remove references to the **four_bit_adder** project in your new **four_bit_add_sub** project, go to the **Project** menu **→ Clean Project →** Check **All revisions →** Click **Ok**.
- After the cleaning process is done, go again to **Project → Revisions →** Double click on **<<new revision>> →** Write **four_bit_add_sub** in the **Revision name** field **→** Leave everything else as they are and click **Ok**.
- Delete the old revision by going back to **Project → Revisions →** Select **four_bit_adder** revision **→** Click **Delete → Yes → Ok**.
- **Re-compile** your project.

With the **four_bit_add_sub.bdf** schematic opened in Quartus, make the needed changes and add necessary logic to the circuit to handle both addition and subtract operation as follows:

- Verify that the four inputs for each of the two 4-bit input numbers (**A[3]..A[0]**, **B[3]..B[0]**) are connected to the first eight switches.
- Change the circuit so that it has an input, using switch number 9, to indicate whether the needed operation is to add or to subtract (0 for add and 1 for subtract). Call this input **Add/Sub**.

- Verify that the four outputs for **Rslt (Rslt[3].. Rslt[0])** are connected to the first four LEDs
- One output for the **Overflow** must be connected to the last LED (LED number 9). You will need to change the overflow digital circuit to handle only signed numbers.

Validate your design on the DE1-SoC board with different inputs. You should try some inputs that generate overflow.

Pack the **four_bit_add_sub** schematic as a logic block named **four_bit_add_sub.bsf** for later use. To do this, go back to your **four_bit_add_sub** schematic and on the **File** main menu, select **Create/Update → Create Symbol Files for Current File** and save the schematic as **four_bit_add_sub.bsf** file.

Assignment 2
Design and implement the 4-bit adder/subtractor in Quartus schematic. Take a screenshot of your design and include it in the lab report. Take pictures of your board for at least 12 test inputs that must include the following cases for both the add and subtract operations:

- All combinations of signed numbers (two positive, two negative, one positive/one negative) for both add and subtract, with and without overflow.

Verify the outputs of your circuit (by converting all input/output numbers to decimal) and explain the results.

Note: to avoid a large report file, make sure to reduce the resolution of the pictures by using a picture management tool that allow saving pictures for print purposes.

4. Lab Submission Instructions

- The lab is intended to be implemented by each **individual** student. It is also allowed for a **maximum of two students** to work in the lab together. If you decide to work in a group of two students, **both students** must **check off** the working lab assignments with a lab instructor and only **one student** must complete the lab submission as explained in the lab assignment on Canvas.
- Write your lab report following the **report template** provided on Canvas. If you are working with a partner, write both names/IDs on the cover page. It is **prohibited** to include a partner's name on a lab report cover page if that partner does not actively participate in implementing the lab.
- The pre-lab questions prepare you for the challenges you will be presented with in the actual lab. Have their answers ready before the lab session and include these answers in the lab report together with the lab assignments.
- For each lab, submit the following on Canvas before the announced due date/time (only submissions on Canvas are accepted):
    - The lab report as a single Word or PDF document
    - Any files the assignments might ask for (e.g., a demonstration video file)
    - Submit each of the above files separately (do not upload compressed files).
- You can submit multiple attempts for this lab; however, only what you submit in the last attempt will be graded (i.e., all required reports and files must be included in this last attempt).
- **Late submission** of any lab assignment after the given due date and time will be penalized at 1 point per every late hour. No assignment submission will be accepted 48 hours after its due date/time.
- **Successful submission**: it is your responsibility to verify that you submitted successfully all the required files for the lab. First, to verify that your files were submitted, go to the assignment details page, and make sure that it says "Successfully submitted" under grade. Second, you need to validate your submission details by clicking the Submission and Rubric link on that page.
- **Deliberate plagiarism** in any assignment is an extremely serious offense that may result in failing the course. Plagiarism includes submitting answers that are not yours (e.g., copied from someone else or from the Internet). Copying materials from the course slides posted on Canvas is not considered plagiarism.